Determinants

Determinants

- If a matrix is square (that is, if it has the same number of rows as columns), then we can assign to it a number called its determinant.

  - Determinants can be used to solve systems of linear equations--as we will see later in the section.

  - They are also useful in determining whether a matrix has an inverse.

- Determinant of a 2 x 2 Matrix

Determinant of 1 x 1 matrix

- We denote the determinant of a square matrix A by the symbol $\det(A)$ or $|A|$.

- We first define $\det(A)$ for the simplest cases.
  - If $A = [a]$ is a 1 x 1 matrix, then $\det(A) = a$.

Determinant of a 2 x 2 Matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

- The determinant of the 2 x 2 matrix

is:

$$\det(A) = |A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

E.g. 1—Determinant of a 2 x 2 Matrix

$$A = \begin{bmatrix} 6 & -3 \\ 2 & 3 \end{bmatrix}$$

- Evaluate $|A|$ for

$$\begin{vmatrix} 6 & -3 \\ 2 & 3 \end{vmatrix} = 6 \cdot 3 - (-3)2 = 18 - (-6) = 24$$

- Determinant of an *n* x *n* Matrix

Determinant of an n x n Matrix

- To define the concept of determinant for an arbitrary *n x n* matrix, we need the following terminology.

Determinant of an n x n Matrix

- Let A be an n x n matrix.

  - The minor M_{ij} of the element a_{ij} is
    the determinant of the matrix obtained by deleting the ith row and jth column of A.

  - The cofactor A_{ij} of the element a_{ij} is:

$$A_{ij} = (-1)^{i+j} M_{ij}$$

Determinant of an n x n Matrix

- For example, A is the matrix $\begin{bmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{bmatrix}$
  - The minor M_{12} is the determinant of the matrix obtained by deleting the first row and second column from A.

$$M_{12} = \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 0 & 4 \\ -2 & 6 \end{vmatrix} = 0(6) - 4(-2) = 8$$

  - So, the cofactor $A_{12} = (-1)^{1+2} M_{12} = -8$

Determinant of an n x n Matrix

- Similarly,

- $$M_{33} = \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix} = 2 \cdot 2 - 3 \cdot 0 = 4$$

- So, $A_{33} = (-1)^{3+3} M_{33} = 4$

Determinant of an n x n Matrix

- Note that the cofactor of a_{ij} is simply
  the minor of a_{ij} multiplied by either 1 or –1, depending on whether $i + j$ is even or odd.

  - Thus, in a 3 x 3 matrix, we obtain the cofactor
    of any element by prefixing its minor with the sign obtained from the following
    checkerboard pattern.

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Determinant of an n x n Matrix

- We are now ready to define
  the determinant of any square matrix.

Determinant of an n x n Matrix

- If A is an n x n matrix, the determinant of A is obtained by multiplying each element of the first row by its cofactor, and then adding the results.

$$\det(A) = |A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

$$= a_{11}A_{11} + a_{12}A_{12} + \ldots + a_{1n}A_{1n}$$

Determinant of a 3 x 3 Matrix

- Evaluate the determinant of the matrix.

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{bmatrix}$$

Determinant of a 3 x 3 Matrix

- $$\begin{aligned}
&\det(A) \\
&= \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{vmatrix} \\
&= 2\begin{vmatrix} 2 & 4 \\ 5 & 6 \end{vmatrix} - 3\begin{vmatrix} 0 & 4 \\ -2 & 6 \end{vmatrix} + (-1)\begin{vmatrix} 0 & 2 \\ -2 & 5 \end{vmatrix} \\
&= 2(2\cdot 6 - 4\cdot 5) - 3\left[0\cdot 6 - 4(-2)\right] - \left[0\cdot 5 - 2(-2)\right] \\
&= -16 - 24 - 4 \\
&= -44
\end{aligned}$$

Expanding the Determinant

- In our definition of the determinant, we used the cofactors of elements in the first row only.
  - This is called expanding the determinant by the first row.
  - In fact, we can expand the determinant by any row or column in the same way, and obtain the same result in each case.
  - We won't prove this, though.

Expanding Determinant about Row and Column

- Let A be the matrix of Example 2.

- Evaluate the determinant of A by expanding

(a) by the second row

(b) by the third column

  - Verify that each expansion gives the same value.

Expanding about Row

- Expanding by the second row, we get:

$$\det(A) = \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & -4 \\ -2 & 5 & 6 \end{vmatrix}$$

$$= -0\begin{vmatrix} 3 & -1 \\ 5 & 6 \end{vmatrix} + 2\begin{vmatrix} 2 & -1 \\ -2 & 6 \end{vmatrix} - 4\begin{vmatrix} 2 & 3 \\ -2 & 5 \end{vmatrix}$$

$$= 0 + 2\left[2 \cdot 6 - (-1)(-2)\right] - 4\left[2.5 - 3(-2)\right]$$

$$= 0 + 20 - 64 = -44$$

Expanding about Column

- Expanding by the third column, we get:

$$\det(A) = \begin{vmatrix} 2 & 3 & -1 \\ 0 & 2 & 4 \\ -2 & 5 & 6 \end{vmatrix}$$

$$= -1\begin{vmatrix} 0 & 2 \\ -2 & 5 \end{vmatrix} - 4\begin{vmatrix} 2 & 3 \\ -2 & 5 \end{vmatrix} + 6\begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix}$$

$$= -[0\cdot 5 - 2(-2)] - 4[2\cdot 5 - 3(-2)]$$

$$+6[2.2 - 3\cdot 0]$$

$$= -4 - 64 + 24 = -44$$

Expanding Determinant about Row and Column

- In both cases, we obtain the same value for the determinant as when we expanded by the first row in Example 2.

Inverse of Square Matrix

- The following criterion allows us to determine whether a square matrix has an inverse without actually calculating the inverse.
  - This is one of the most important uses of the determinant in matrix algebra.
  - It is reason for the name determinant.

Invertibility Criterion

- If A is a square matrix, then A has an inverse if and only if $\det(A) \neq 0$.

  - We will not prove this fact.
  - However, from the formula for the inverse of a 2 x 2 matrix, you can see why it is true in the 2 x 2 case.

Determinant to Show Matrix Is Not Invertible

- Show that the matrix A has no inverse.

$$A = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 3 \\ 5 & 6 & 2 & 6 \\ 2 & 4 & 0 & 9 \end{bmatrix}$$

  - We begin by calculating the determinant of A.
  - Since all but one of the elements of the second row is zero, we expand the determinant by the second row.

Determinant to Show Matrix Is Not Invertible

- If we do so, we see from this equation that only the cofactor A_{24} needs to be calculated.

$$\det(A)$$

$$= \begin{vmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 0 & 3 \\ 5 & 6 & 2 & 6 \\ 2 & 4 & 0 & 9 \end{vmatrix}$$

$$= -0 \cdot A_{21} + 0 \cdot A_{22} - 0 \cdot A_{23} + 3 \cdot A_{24}$$

$$= 3A_{24}$$

Determinant to Show Matrix Is Not Invertible

$$= 3\begin{vmatrix} 1 & 2 & 0 \\ 5 & 6 & 2 \\ 2 & 4 & 0 \end{vmatrix}$$

$$= 3(-2)\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}$$

$$= 3(-2)(1 \cdot 4 - 2 \cdot 2)$$

$$= 0$$

- Since the determinant of A is zero, A cannot have an inverse—by the Invertibility Criterion.

- Row and Column Transformations

Row and Column Transformations

- The preceding example shows that, if we expand a determinant about a row or column that contains many zeros, our work is reduced considerably.
  - We don't have to evaluate the cofactors of the elements that are zero.

Row and Column Transformations

- The following principle often simplifies the process of finding a determinant by introducing zeros into it without changing its value.

Row and Column Transformations of a Determinant

- If A is a square matrix, and if the matrix B is obtained from A by adding a multiple of one row to another, or a multiple of one column to another, then

$$\det(A) = \det(B)$$

Using Row and Column Transformations

- Find the determinant of the matrix A.

$$A = \begin{bmatrix} 8 & 2 & -1 & -4 \\ 3 & 5 & -3 & 11 \\ 24 & 6 & 1 & -12 \\ 2 & 2 & 7 & -1 \end{bmatrix}$$

  - Does it have an inverse?

Using Row and Column Transformations

- If we add –3 times row 1 to row 3, we change all but one element of row 3 to zeros:

$$\begin{bmatrix} 8 & 2 & -1 & -4 \\ 3 & 5 & -3 & 11 \\ 0 & 0 & 4 & 0 \\ 2 & 2 & 7 & -1 \end{bmatrix}$$

  - This new matrix has the same determinant as *A*.

Using Row and Column Transformations

- If we expand its determinant by the third row, we get:

$$\det(A) = 4\begin{vmatrix} 8 & 2 & -4 \\ 3 & 5 & 11 \\ 2 & 2 & -1 \end{vmatrix}$$

  - Now, adding 2 times column 3 to column 1 in this determinant gives us the following result.

Using Row and Column Transformations

$$\det(A) = 4\begin{vmatrix} 0 & 2 & -4 \\ 25 & 5 & 11 \\ 0 & 2 & -1 \end{vmatrix} = 4(-25)\begin{vmatrix} 2 & -4 \\ 2 & -1 \end{vmatrix}$$

$$= 4(-25)\left[2(-1) - (-4)2\right]$$

$$= -600$$

- Since the determinant of *A* is not zero, *A* does have an inverse.

Example 1:

We can compute the determinant
$$|\mathbf{T}| = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$$

by expanding along the first row,

$$|\mathbf{T}| = 1\times(-)^{1+1}\begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} + 2\times(-)^{1+2}\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3\times(-)^{1+3}\begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} \quad = -3+12-9=0$$

Or expand down the second column:

$$|\mathbf{T}| = 2\times(-)^{1+2}\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 5\times(-)^{2+2}\begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} + 8\times(-)^{3+2}\begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} \quad = 12-60+48=0$$

Example 2: (using a row or column with many zeroes)

$$\begin{vmatrix} 1 & 5 & 0 \\ 2 & 1 & 1 \\ 3 & -1 & 0 \end{vmatrix} = 1\times(-)^{2+3}\begin{vmatrix} 1 & 5 \\ 3 & -1 \end{vmatrix} \quad = 16$$

Properties of determinants

Property 1: If one row of a matrix consists entirely of zeros, then the determinant is zero.

$$\text{We have } \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 0 & 0 & 0 \end{vmatrix} = 0.$$

Property 2: If two rows of a matrix are interchanged, the determinant **changes sign**.

$$\text{We have } |A| = \begin{vmatrix} 2 & -1 \\ 3 & 2 \end{vmatrix} = - \begin{vmatrix} 3 & 2 \\ 2 & -1 \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ -1 & 2 \end{vmatrix} = |A^T| = 7.$$

Properties of determinants

Property 1: If one row of a matrix consists entirely of zeros, then the determinant is zero.

Property 2: If two rows of a matrix are interchanged, the determinant **changes sign**.

Property 3: If two rows of a matrix are identical, the determinant is zero.

We have $\begin{vmatrix} 1 & 2 & 3 \\ -1 & 0 & 7 \\ 1 & 2 & 3 \end{vmatrix} = 0.$

Properties of determinants

Property 1: If one row of a matrix consists entirely of zeros, then the determinant is zero.

Property 2: If two rows of a matrix are interchanged, the determinant **changes sign**.

Property 3: If two rows of a matrix are identical, the determinant is zero.

Property 4: If the matrix **B** is obtained from the matrix **A** by multiplying every element in one row of **A** by the scalar λ, then |**B**|= λ|**A**|.

We have $\begin{vmatrix} 2 & 6 \\ 1 & 12 \end{vmatrix} = 2\begin{vmatrix} 1 & 3 \\ 1 & 12 \end{vmatrix} = (2)(3)\begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix} = 6(4-1) = 18.$

We have

$$\begin{vmatrix} 1 & 2 & 3 \\ 1 & 5 & 3 \\ 2 & 8 & 6 \end{vmatrix} = 2 \begin{vmatrix} 1 & 2 & 3 \\ 1 & 5 & 3 \\ 1 & 4 & 3 \end{vmatrix} = (2)(3) \begin{vmatrix} 1 & 2 & 1 \\ 1 & 5 & 1 \\ 1 & 4 & 1 \end{vmatrix} = (2)(3)(0) = 0.$$

Here, we first factored out 2 from the third row and 3 from the third column, and then used Theorem 3.3, since the first and third columns are equal. ■

Properties of determinants

Property 1: If one row of a matrix consists entirely of zeros, then the determinant is zero.

Property 2: If two rows of a matrix are interchanged, the determinant **changes sign**.

Property 3: If two rows of a matrix are identical, the determinant is zero.

Property 4: If the matrix **B** is obtained from the matrix **A** by multiplying every element in one row of **A** by the scalar λ, then $|\mathbf{B}| = \lambda|\mathbf{A}|$.

Property 5: For an $n \times n$ matrix **A** and any scalar λ, $\det(\lambda\mathbf{A}) = \lambda^n \det(\mathbf{A})$.

Property 6: If a matrix **B** is obtained from a matrix **A** by adding to one row of **A**, a scalar times another row of **A**, then |**A**|=|**B**|.

We have

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & -1 & 3 \\ 1 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 5 & 0 & 9 \\ 2 & -1 & 3 \\ 1 & 0 & 1 \end{vmatrix},$$

obtained by adding twice the second row to the first row. By applying the definition of determinant to the first and second determinant, both are seen to have the value 4. ■

Property 6: If a matrix **B** is obtained from a matrix **A** by adding to one row of **A**, a scalar times another row of **A**, then |**A**|=|**B**|.

Property 7: det(**A**) = det(**A**^{T}).

Property 8: The determinant of a triangular matrix, either upper or lower, is the product of the elements on the main diagonal.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -8 & -4 \\ 0 & 0 & -\frac{30}{4} \end{bmatrix}$$

$$\det(A) = (1)(-8)\left(-\frac{30}{4}\right)$$

Property 6: If a matrix **B** is obtained from a matrix **A** by adding to one row of **A**, a scalar times another row of **A**, then $|\mathbf{A}|=|\mathbf{B}|$.

Property 7: $\det(\mathbf{A}) = \det(\mathbf{A}^T)$.

Property 8: The determinant of a triangular matrix, either upper or lower, is the product of the elements on the main diagonal.

Property 9: If A and B are of the same order, then

$$\det(AB)=\det(A)\,\det(B).$$

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

Then

$$|A| = -2 \quad \text{and} \quad |B| = 5.$$

On the other hand, $AB = \begin{bmatrix} 4 & 3 \\ 10 & 5 \end{bmatrix}$, and $|AB| = -10 = |A||B|$.

- Find the determinant of $A = \begin{bmatrix} 2 & -3 & 10 \\ 1 & 2 & -2 \\ 0 & 1 & -3 \end{bmatrix}$

$$\begin{vmatrix} 2 & -3 & 10 \\ 1 & 2 & -2 \\ 0 & 1 & -3 \end{vmatrix} = -\begin{vmatrix} 1 & 2 & -2 \\ 2 & -3 & 10 \\ 0 & 1 & -3 \end{vmatrix} \times(-2) = -\begin{vmatrix} 1 & 2 & -2 \\ \boxed{0 \quad -7 \quad 14} \\ 0 & 1 & -3 \end{vmatrix}$$

Factor -7 out of the 2nd row

$$= 7\begin{vmatrix} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & 1 & -3 \end{vmatrix} \times(-1) = 7\begin{vmatrix} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & -1 \end{vmatrix} = 7(1)(1)(-1) = -7$$

We can also obtain $|A|$ as follows. Repeat the first and second columns of A, as shown next. Form the sum of the products of the entries on the lines from left to right, and subtract from this number the products of the entries on the lines from right to left (verify):

$$\begin{array}{ccccc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array} .$$

■

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 1 & 2 \end{bmatrix}.$$

Evaluate $|A|$.

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 1 & 2 \end{bmatrix}.$$

Evaluate $|A|$.

Solution

Substituting in (1), we find that

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 1 & 2 \end{vmatrix} = (1)(1)(2) + (2)(3)(3) + (3)(2)(1) - (1)(3)(1) - (2)(2)(2) - (3)(1)(3) = 6.$$

We could obtain the same result by using the easy method illustrated previously, as follows:

$$\begin{array}{ccccc} 1 & 2 & 3 & 1 & 2 \\ 2 & 1 & 3 & 2 & 1 \\ 3 & 1 & 2 & 3 & 1 \end{array}$$

$$|A| = (1)(1)(2) + (2)(3)(3) + (3)(2)(1) - (3)(1)(3) - (1)(3)(1) \\ - (2)(2)(2) = 6.$$ ■

Warning The methods used for computing $\det(A)$ in Examples

do not apply for $n \geq 4$.

Corollary

If A is nonsingular, then $\det(A^{-1}) = \dfrac{1}{\det(A)}$.

Corollary

If A and B are similar matrices, then $\det(A) = \det(B)$.

Be Careful!!!!

The determinant of a sum of two $n \times n$ matrices A and B is, in general, not the sum of the determinants of A and B. The best result we can give along these lines is that if A, B, and C are $n \times n$ matrices all of whose entries are equal except for the kth row (column), and the kth row (column) of C is the sum of the kth rows (columns) of A and B, then $\det(C) = \det(A) + \det(B)$. We shall not prove this result, but will consider an example.

Let

$$A = \begin{bmatrix} 2 & 2 & 3 \\ 0 & 3 & 4 \\ 0 & 2 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 2 & 3 \\ 0 & 3 & 4 \\ 1 & -2 & -4 \end{bmatrix},$$

and

$$C = \begin{bmatrix} 2 & 2 & 3 \\ 0 & 3 & 4 \\ 1 & 0 & 0 \end{bmatrix}.$$

Then $|A| = 8$, $|B| = -9$, and $|C| = -1$, so $|C| = |A| + |B|$.

Let A be a 4×4 matrix with $\det(A) = -2$.

(a) Describe the set of all solutions to the homogeneous system $A\mathbf{x} = \mathbf{0}$.

(b) If A is transformed to reduced row echelon form B, what is B?

(c) Give an expression for a solution to the linear system $A\mathbf{x} = \mathbf{b}$, where

$$\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}.$$

(d) Can the linear system $A\mathbf{x} = \mathbf{b}$ have more than one solution? Explain.

(e) Does A^{-1} exist?

Solution

(a) Since $\det(A) \neq 0$, the homogeneous system has only the trivial solution.

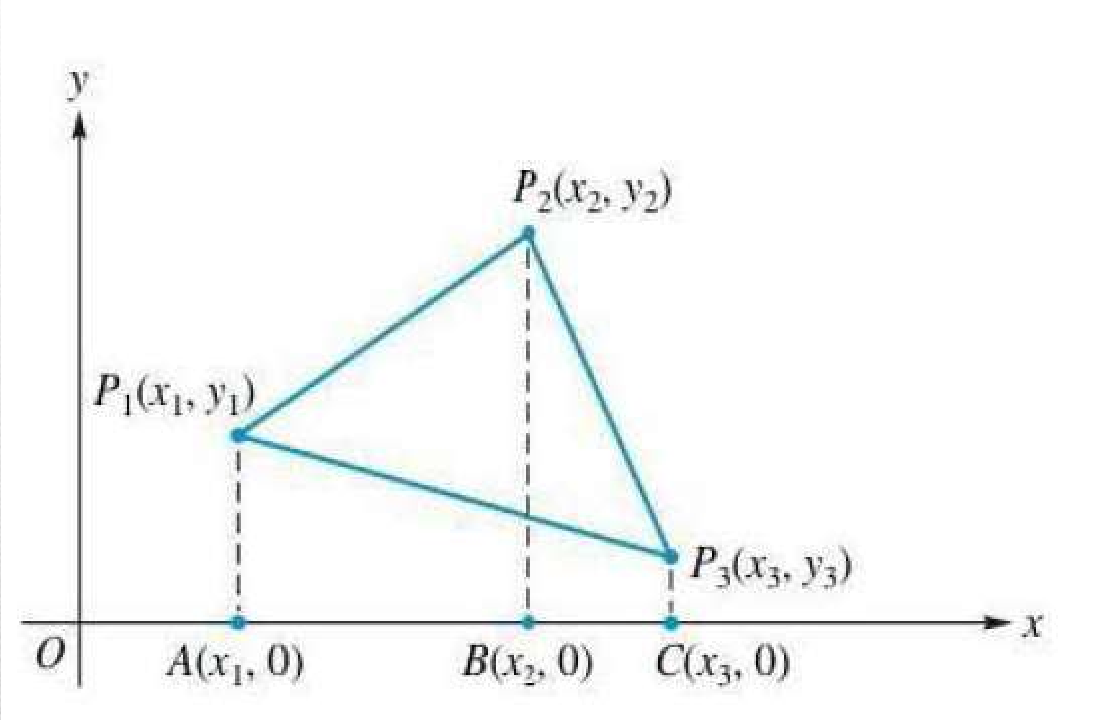
(b) Since $\det(A) \neq 0$, A is a nonsingular matrix, $B = I_n$.

(c) A solution to the given system is given by $\mathbf{x} = A^{-1}\mathbf{b}$.

(d) No. The solution given in part (c) is the only one.

(e) Yes.

Application to Computing Areas


y
$P_2(x_2, y_2)$
$P_1(x_1, y_1)$
$P_3(x_3, y_3)$
x
O
$A(x_1, 0)$
$B(x_2, 0)$
$C(x_3, 0)$

$$\text{area of triangle} = \frac{1}{2}\left|\det\left(\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}\right)\right|.$$

Compute the area of the triangle T, shown in Figure 3.3, with vertices $(-1, 4)$, $(3, 1)$, and $(2, 6)$.

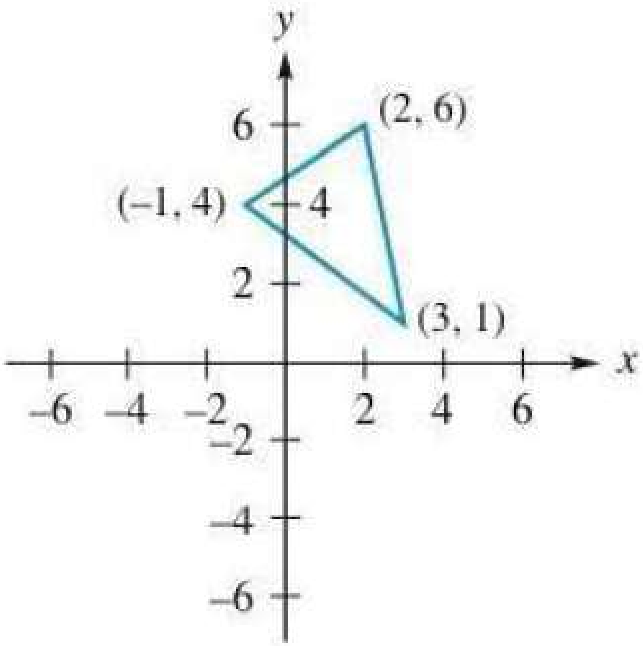


FIGURE 3.3

Solution

By Equation (3), the area of T is

$$\frac{1}{2}\left|\det\left(\begin{bmatrix} -1 & 4 & 1 \\ 3 & 1 & 1 \\ 2 & 6 & 1 \end{bmatrix}\right)\right| = \frac{1}{2}|17| = 8.5. \qquad \blacksquare$$